SAGAR RAJAK

Backend & Agentic AI Engineer

Mumbai, India | GitHub | Portfolio | LinkedIn | sagarrajak245@gmail.com | +91-8669101264

SUMMARY

Backend and Agentic AI Engineer building production systems end-to-end from event-driven distributed backends to multi-agent LLM platforms. Founding engineer on an AI product now used by paying enterprise clients; designed fault-tolerant, at-least-once processing pipelines and a real-time voice AI stack from scratch.

EXPERIENCE

Aiviue (Founding Team) — *Agentic AI Engineer* **Nov 2025 – Present**

- Owned ~90% of the MVP codebase (architecture, backend infrastructure, agentic workflows) as founding engineer on an AI video-generation hiring platform; shipped the product that landed **Hitachi, Orange Antelopes, and Tapplent** as paying clients.
- Designed an event-driven backend (FastAPI, PostgreSQL, Redis Streams) with idempotent consumers and dead-letter queues, achieving reliable at-least-once processing with zero data-loss incidents across production traffic.
- Built a 6-agent control plane (cost attribution, anomaly detection, predictive analytics, automated remediation) across multiple tenants, cutting manual ops intervention and enabling real-time cost/profitability visibility.
- Engineered multi-channel SDR automation (LinkedIn, Email, WhatsApp) into a single pipeline, reducing manual marketing effort by **60%** via AI-driven enrichment and webhook-secured tracking.
- Replaced third-party voice orchestration with an in-house stack (multilingual STT/TTS, LangGraph agent workflows, real-time telephony), delivering low-latency, fault-tolerant voice AI interactions in production.

Kinzy Club Pvt Ltd (Subsidiary of Redscope Inc.) — *Backend Engineer* **Nov 2024 – Feb 2025**

- Built secure JWT-based authentication with interceptors and route protection for a high-traffic service.
- Eliminated sequential scans on 42 high-traffic queries via targeted indexing, cutting average query latency from **10ms to 2ms**.
- Built a Redis-backed feed system using partial active indexing to reduce database load.
- Scheduled distributed tasks with Spring Scheduler + ShedLock, using per-node locks to prevent duplicate execution across nodes.
- Wrote unit and integration tests (JUnit, MockMvc, Spring Test) ensuring reliable request handling and regression safety.

AI-Colegion — *Technical Officer* **Aug 2023 – May 2026**

- Led 3 MERN-stack club projects and instructed 350+ students across Flutter, Machine Learning, and Agentic AI (CrewAI) workshops.

PROJECTS

Postify — AI-Powered Chrome Extension [github]

React, TypeScript, Tailwind CSS, GPT-4 / Grok / Gemini, Gmail API

- Built a productivity-focused Chrome Extension that automates certificate discovery and social sharing, removing a manual, multi-app workflow for personal branding.
- Integrated Gmail API with Google OAuth2 to securely scan and parse certificate emails, extracting course metadata (issuer, title, completion date) without manual data entry.
- Generated ready-to-post LinkedIn and Twitter copy by routing parsed metadata through GPT-4, Grok, and Gemini, letting the user pick the strongest draft across providers.
- Implemented smart scheduling and one-click publishing across Gmail, LinkedIn, and Twitter APIs, turning a certificate into a published post in under a minute end-to-end.

IntelliClaim — Multi-Agent AI System for Insurance Adjudication [github]

React, Flask, CrewAI, LangChain, RAG, NLP, LLMs

- Architected a full-stack, multi-agent system that automates insurance claim adjudication, replacing a manual document-review process with a coordinated agent pipeline.
- Designed a team of specialized CrewAI + LangChain agents (intake, policy analysis, decisioning) powered by Groq's high-speed inference for low-latency reasoning.
- Built an intelligent document-retrieval layer using Retrieval-Augmented Generation with ChromaDB and spaCy-based NLP to parse and query complex, unstructured policy documents.
- Built a real-time Socket.io dashboard that visualizes each agent's reasoning step live, giving adjusters explainable-AI (XAI) visibility into why a claim was approved or flagged.

WebRTC Real-Time Multi-Object Detection [github]

WebRTC, YOLOv8, ONNX (WASM), Node.js, FastAPI, Docker

- Built a real-time object-detection system that streams live video from a mobile device over WebRTC, runs YOLOv8 inference, and overlays detections with minimal latency.
- Implemented dual inference modes — in-browser WASM for lightweight, on-device detection and a server mode for high-accuracy processing — trading off accuracy vs. device load depending on use case.
- Benchmarked latency, FPS, and CPU usage across both modes, achieving **<100ms latency in server mode** and **10–30 FPS** across modes.
- Containerized the full pipeline with Docker and integrated ngrok for external phone access, enabling fast, repeatable real-time testing without local network constraints.

EDUCATION

Vivekanand Education Society's Institute of Technology, Mumbai Jan 2026

Bachelor of Engineering in Artificial Intelligence and Data Science — CGPA: 9.14/10

MHT-CET — 97.25 Percentile

SKILLS

Languages: Python, JavaScript, SQL, C/C++, Java

AI / Agentic: LangChain, LangGraph, CrewAI, RAG, PyTorch, Machine Learning (Supervised & Unsupervised ML), CNN, RNN ,Transformers, Context Engineering.

AI Observability: LangFuse

Backend & Infra: Distributed System, FastAPI,Docker, Redis, Redis Streams, Git, GitHub Actions (CI/CD), Jenkins

Cloud & Monitoring: AWS (EC2, S3, Lambda), Prometheus, Grafana

Databases: PostgreSQL, MongoDB, ChromaDB, pgvector (Vector DBs)

Frontend: React js, Next js

CERTIFICATIONS

AWS Cloud Fundamentals · NVIDIA: Fundamentals of Deep Learning